Data-driven *pricing*

Starting with luxury event venues, where every date is a one-off and every price is a negotiation.

01 THE PARTNER

A Manhattan event venue selling event days at

$50K–$200K

Galas, conferences and celebrations, booked one date at a time.

02 THE INCUMBENT

A static rate card

Varies only by year, month and weekday. It never sees:

- Event type
- Lead time
- Demand pressure

Every quote is negotiated by hand. The hardest call, *take this deal or hold the date for a better one*, is made on gut feel.

03 MLIQ

Replace the card with models trained on the venue's own sales history.

Delivered inside the existing Salesforce workflow, so pricing and lead scoring live where the sellers already work.

Predict, then *prescribe*

Stage 1 predicts what shows up and what it's worth. Stage 2 prescribes the call.

PREDICTIVE · MODEL 1 · DEMAND

What kind of demand does this date attract?

Event type and calendar footprint. A conference books midweek and consumes 2–4 days with load-ins; a gala takes one.

PREDICTIVE · MODEL 2 · VALUE

What should this date cost?

By event type, section, weekday and month, as a **negotiable range**, not a single number.

PRESCRIPTIVE · PRICING THE OPPORTUNITY COST

Accepting a one-day social on a Wednesday destroys the **Wed+Thu** and **Tue+Wed** conference blocks, the two most common multi-day shapes in the book.

$195K

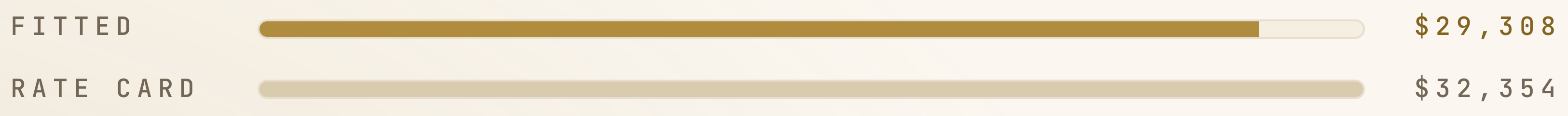
MEDIAN BLOCK VALUE

ONE DECISION RULE

`Hold the date if` `P(better arrives)` × `E[value]` − `holding cost` > `the offer in hand`

THE BAR TO CLEAR · MAE, LOWER IS BETTER

56.4% HEAD-TO-HEAD VS THE RATE CARD



Curves fitted to realized bookings already beat the card, but only *in-sample today*. The bake-off re-measures it on an out-of-time holdout. **The models must clear the fitted curves, not the card.**

One venue's history, *2021–2026*

Extracted and anonymized end-to-end: pseudonyms, an allow list of approved fields, and no names, emails or free text leaving the CRM.

3,985
OPPORTUNITIES · **THE DEALS**

5,877
RESERVATIONS · **THE CALENDAR**

1,170
LOST OPPORTUNITIES **RESTORED**

$131M
QUOTED-AND-LOST REVENUE **RECOVERED IN CLEANUP**

THE HOLE

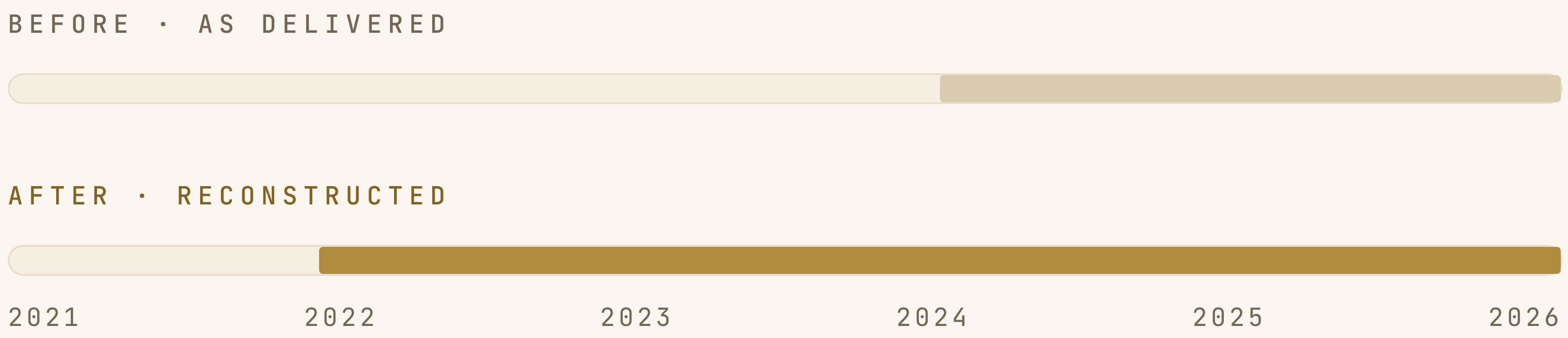
2021–2023 was migrated from a legacy CRM *as wins only*.

The lost deals existed as empty shells, or not at all. So the file remembered every price that worked and forgot every price that didn't — and a model trained on that learns **what we charged**, never **what the market refused**.

266
REBUILT FROM SCRATCH

904
EMPTY SHELLS REPAIRED

USABLE DEMAND HISTORY



Demand history now runs from **2022**, not 2024. **The modeling window doubled.**

Which matters because pricing is seasonal: four booking years lets us see each season repeat, instead of guessing a pattern from one pass through the calendar.

Finding the *grain*

The unit of analysis had to be earned. The calendar lies about money.

THE TRAP

Reservation money fields are formulas that read the **parent Opportunity**. A three-day booking doesn't split its value across the three days — it *repeats the full deal amount on each one*. So summing day money off the calendar inflates revenue

2.5×

THE FIX

Rebuild per-day dollars from the **product lines underneath** — the itemized charges on the contract, which are booked once each rather than repeated. They reconcile to opportunity totals to the cent.

Reconciled to the cent

THE TARGET

Venue rent per event day

$80,000

MEDIAN WON · FULL VENUE $90K · PARTIAL $34.75K

MODELING GRAIN 1

3,706

EVENTS · ONE PER INQUIRY: OUTCOME, EVENT TYPE, TRUE INQUIRY AND LOSS DATES

MODELING GRAIN 2

5,851

EVENT-DAYS · ONE PER DAY PER VENUE SECTION: ITS OWN HOLD POSITION, ITS OWN DOLLARS

OUTCOMES

659	2,981	66
WON	LOST	OPEN

4.5:1 lost to won. Exactly why a model that learns only from wins **misreads the market**.

Inquiry to *Booked*

Every inquiry lands as a Lead. Once qualified it becomes an Opportunity that holds dates on the master calendar.

01 CAPTURE · ANY CHANNEL, ONE ENTRY POINT

CHANNELS

Webform · Email · Phone · Marketplace

→

OBJECT

Lead

→

CHILD OF LEAD

Lead Product

Venue rent, beverage, staffing

CHILD OF LEAD

Lead Reservation

Requested dates + venue sections

Each Lead Reservation looks up **one** Lead Product, and the Lead.

02 QUALIFY · SALES REVIEWS EVERY LEAD

DECISION

Lead review

→

QUALIFIED

Account · Contact · Opportunity

Lead Products and Lead Reservations carry over

UNQUALIFIED

Marked and closed. No Account, Contact or Opportunity created

Pipeline, holds and *money*

The money rides on the products attached to each day.

03 PIPELINE · OPPORTUNITY STAGES

STAGE 1	STAGE 2	STAGE 3	STAGE 4	WON HERE	DELIVERED
Qualified	Proposal	Negotiation	Contract	Booked	Closed Won

Won at Booked. Contract signed, deposit in. Closed Won is bookkeeping after delivery and deposit refund.

Closed Lost. Exit from any stage, mostly on *date availability* and *price*.

04 CALENDAR AND MONEY

MASTER CALENDAR

Reservation

Each product on the Opportunity, every day, load-in included

CHILD OF OPPORTUNITY

Opportunity Product

Venue rent · Beverage · Staffing. One row per item, per day

Per-day pricing. Money sits on each day's line items, not the Opportunity as a whole.

HOLD POSITION · ONE DATE, MANY DEALS

1st	First claim on the date
2nd	Moves up if the 1st is released
3rd	Still tracked and quotable

WIN

Holds flip to **Confirmed**; date blocked.

LOSS

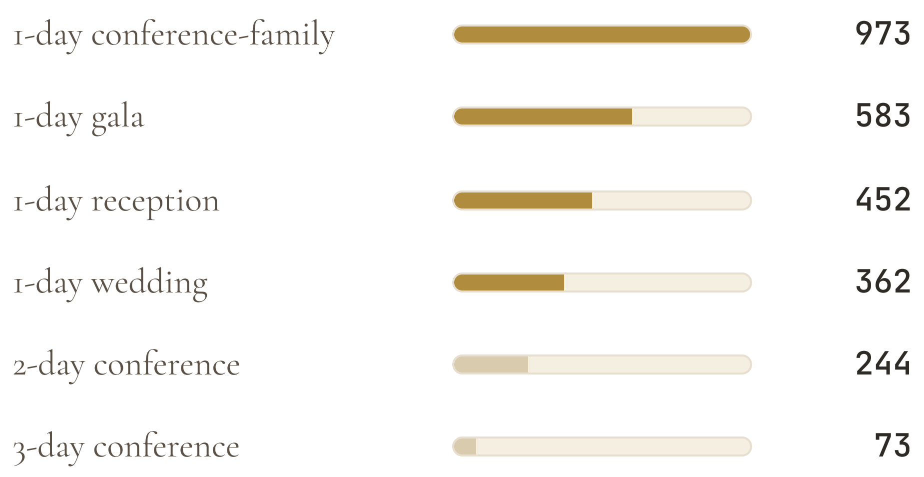
Date released; **next hold moves up**.

The two *predictive* models

Different grains, different training sets.
One decision.

	MODEL 1 Demand	MODEL 2 Value
ESTIMATES	Arrival rate **λ** per demand shape, by weekday, month and days-out. **P(better deal in k days)**	Venue rent per event day, as a **low / point / high** range, because every price here is negotiated.
ALGORITHM	Poisson arrival model, partial pooling across sparse shape cells	Gradient-boosted quantile regression on log rent/day
GRAIN	**Event (3,706)**. An event arrives once, then occupies days; day grain would count a 3-day conference three times	**Event-day (5,851)**. The same event's Thursday and Saturday are worth different money
FEATURES	weekday, month, lead time, demand shape (type × footprint), market segment	event type, venue section, weekday, month, guest count, lead time, footprint length, segment, industry
TRAINS ON	All arrivals, **won AND lost**. A deal lost to a blocked date is displaced demand, not a price rejection	Priced won days + clean rejections only; blocked-date losses excluded, because their high quotes would teach *"high prices lose"*

THE DEMAND SHAPES ARE REAL CELLS, NOT THEORY



Thin tails share strength through partial pooling; the **multi-day corporate shapes** are the $195K blocks the opportunity-cost math protects.

ONE SHARED TRAIN / SERVE CONTRACT

9 required + **5 optional** features, with leakage forbidden by construction: rules-engine outputs, post-outcome fields and dead source fields **raise an error** if they reach a model.

Alternatives

Three we considered. Same data, same test, scored once — so nothing wins on reputation.

ALTERNATIVE	GRAIN	VERDICT
Multinomial event-type model Advisor's suggestion: which *kind* of event is likely to want a given date	Event	**ADOPTED** It tells us *what* books a date. It can't tell us *how many* ask at all — so that stays a separate model. A slow month isn't the same thing as a change in client mix.
Summed one-day opportunity costs Price a hold as the sum of individual day costs	Event-day	**KEPT AS A FLOOR** It prices too low: one Wednesday booking kills *every* multi-day block crossing Wednesday. So a lost block is priced **whole, not day by day**.
Neural networks One net for price, one for arrivals	Same as the champion each challenges	**RUN AS CHALLENGERS** Only worth trying because the rebuild doubled our history. Same rows, same splits — they win the holdout or they don't ship.

ONE PARTITION DESIGN · OUT-OF-TIME 60 / 20 / 20

TRAIN 60 · VALIDATE 20 · HOLDOUT 20

Split by date, so the future never trains the model on itself. Grouped by client, so **nobody trains it and then grades it**. The last 20% is held back and **scored once**.

WHY GBM STARTS CHAMPION

On ~3,700 tabular events: it quotes a *range*, not one number; it respects rules the business already knows; and it shows what moved the price.

Whatever wins must also beat the re-fit curves on the holdout. **Nothing ships on architecture appeal.**

What we're claiming, and what would *prove us wrong*

The data work is done. What's left is the measurement we've already committed to.

01 THE CLAIM

A model trained on this venue's own history prices a date better than a card that cannot see event type, lead time or demand pressure.

And it prices the harder question the card never asks: *hold, or take the offer in hand?*

02 THE PROOF

One out-of-time holdout, scored once.

Champions must beat the **re-fit curves**, not the rate card. Beating the card is the floor, not the result.

No leakage by construction; nothing promoted on architecture appeal.

03 THE FAILURE WE ALREADY PRICED

A model that learns only from wins.

4.5:1 lost to won is why the 1,170 lost opportunities were rebuilt *before* anything was fit.

A blocked-date loss is displaced demand, not a price rejection. The two models split on exactly that line.

NEXT · 01

Score the out-of-time holdout **once**, and report whether the models clear the re-fit curves.

NEXT · 02

Run the challengers on identical rows and partitions. Promote only on a **shared-holdout win**.

NEXT · 03

Put the quote and the hold call in front of sellers **inside Salesforce**, where the work already happens.